

renewable energy sector still has a significant level of import dependence, particularly in solar cells, and the critical upstream segment, especially wafers, ingots, and polysilicon, remain the next major focus area for the industry.

“Reducing import dependence will be critical for ensuring long-term energy security, supply chain stability, and cost competitiveness as India moves towards its ambitious clean energy targets. India’s renewable energy ambitions are scaling rapidly, and manufacturing capabilities must grow alongside deployment,” says Thadani.

IMPORT RELIANCE

The reliance is primarily on China, which supplies 95% of the world’s wafers and polysilicon. Without rapid upstream investment, the supply-demand gap for these inputs will widen as installation volumes grow.

Closing this gap will require large capital expenditures for vertical integration. Manufacturing polysilicon and wafers is capital-intensive, with long lead times and high technological barriers. To achieve an additional 30–40 GW of upstream capacity (cells, wafers, polysilicon) by 2030, tens of billions of dollars will need to be invested.

“Industry estimates suggest setting up a modern integrated polysilicon-wafer-cell production facilities can cost \$150–350 million per GW of annual capacity depending on the technology used. Fully integrated polysilicon-to-module facilities may range from \$200–400 million per GW, reflecting additional module production costs and India-specific infrastructure challenges,” says Aniket Tiwari, Research Associate, TERI.

TOP PV MANUFACTURING COUNTRIES’ MARKET SHARE

- Huge dependence on China for manufacturing
- India has no capacity for wafers and polysilicon

COUNTRY	MODULE	CELLS	WAFERS	POLYSILICON
China	84.6	91.8	98	92
Vietnam	3.4	1.7	2	0
India	2.7	0.5	0	0
Thailand	2.3	0.6	0	0
US	2.2	0.8	0	2
Malaysia	2.1	2.3	0	2
Germany	0	0	0	4
Rest of the World	2.6	2	0	0

NOTE FIGURES IN %; SOURCE IEA

DOMESTIC SUPPLY VS DEMAND FOR PV VALUE CHAIN

- India’s dependence on China to continue for wafers/cells
- Need to explore the market for module exports

VALUE CHAIN	MANUFACTURING CAPACITY (GW/YEAR)	DEMAND FOR 2030 (GW/YEAR)	SURPLUS/DEFICIT (-) (GW)/YEAR	SUPPLY-DEMAND GAP ANALYSIS
Module	200	54	146	○ Surplus; various export channels need to be explored
Cell	27	54	-27	○ With current planned capacities, the import reliance will be reduced
Ingots/Wafers	2	52	-52	○ 90% of wafers still need to be imported; at least 30 GW of new slicing capacity is required before 2028
Polysilicon	0	54	-51	○ 100% reliance on foreign supply (mainly China). Two announced megapoly plants (Adani, Reliance) must be commissioned on time

SOURCE TERI/MNRE